

PROJECT DOCUMENTATION: RETAIL SALES ANALYTICS USING EXCEL, SQL, POWER BI, AND PYTHON

1. Introduction

In today's highly competitive retail landscape, businesses face significant challenges in tracking comprehensive sales performance, understanding nuanced customer behavior, and identifying clear opportunities for growth. This project delivers an end-to-end analytics pipeline designed to transform raw transactional data into actionable business intelligence. By leveraging a multi-tool approach—spanning data cleaning, relational databases, machine learning, and interactive visualization—this solution empowers management to make data-driven decisions regarding sales, inventory, and targeted marketing strategies.

2. Business Objectives & Use Cases

The core objective of this project is to build a robust, integrated analytics solution that addresses the following key business use cases:

- Analyzing sales performance across various stores and individual staff members.
- Identifying high-value customers and mapping their buying behavior patterns.
- Performing advanced customer segmentation to enable targeted marketing campaigns.
- Monitoring active inventory levels and tracking product performance to prevent stockouts.
- Deploying an interactive Power BI dashboard to provide real-time insights for management.

3. Technology Stack & Domain

- **Domain:** Retail, Sales & Inventory Analytics
- **Data Cleaning & Preprocessing:** Microsoft Excel
- **Data Storage, Querying & Transformation:** SQL
- **Exploratory Data Analysis & Machine Learning:** Python (Pandas, NumPy, Matplotlib, Seaborn, KMeans/Clustering)

- **Business Intelligence & Reporting:** Microsoft Power BI

4. Step-by-Step Implementation Pipeline

Data Cleaning & Preparation (Excel)

The project began by establishing data hygiene across the provided retail datasets (customers, orders, order_items, products, brands, categories, stocks, stores, and staffs).

- **Standardization:** Applied consistent naming conventions (e.g., snake_case) to all column headers across sheets.
- **Quality Control:** Removed duplicate records and addressed missing values by filling blanks (such as ZIP codes and phone numbers) with "NA" placeholders.
- **Data Formatting & Validation:** Converted relevant columns to standard Excel Date and Number formats, and implemented dropdown validations for fields like order_status.
- **Feature Engineering:** Derived a new total_price column using the formula (list_price * quantity) - discount and merged product names into order items via VLOOKUP/XLOOKUP.
- **Export:** Saved the finalized, clean datasets as CSV files ready for database ingestion.

Database Management & Querying (SQL)

A relational database was constructed to house the cleaned data, allowing for robust querying and aggregation.

- **Schema Design:** Created SQL tables replicating the exact structure and constraints of the project's Entity-Relationship Diagram (ERD).
- **Data Ingestion:** Imported the cleaned CSV files into the SQL database.
- **Core Querying:** Executed INNER JOIN operations connecting orders, order_items, and products to extract detailed line-item views.
- **Business Logic:** Developed queries to group total sales by store_id, identify the top 5 selling products by quantity, analyze staff revenue performance, and calculate total customer spend.

- **Operational Alerts:** Created a stock alert query to list products with an inventory of fewer than 10 units in any store.
- **Preparation for ML:** Generated an empty customer_segments table to later store machine learning outputs.

Advanced Analytics & Machine Learning (Python)

Python was utilized to perform deep exploratory data analysis and segment the customer base using machine learning.

- **Database Connection:** Leveraged `pandas.read_sql()` to directly query and load SQL tables (orders, order_items, customers) into Python DataFrames.
- **Exploratory Data Analysis (EDA):** Summarized datasets using `df.describe()` and `df.info()`, and visualized baseline distributions using Matplotlib and Seaborn.
- **RFM Modeling:** Computed Recency (days since last order), Frequency (number of total orders), and Monetary (total value of all purchases) features for every customer.
- **Clustering:** Applied KMeans Machine Learning to categorize customers into distinct segments based on their calculated RFM scores.
- **Data Pipeline Completion:** Exported the resulting segmentation labels back into the SQL customer_segments table.

Visualization & BI Dashboarding (Power BI)

The final phase synthesized the relational data and machine learning outputs into an interactive dashboard.

- **Data Modeling:** Connected Power BI directly to the SQL database, establishing foreign key relationships across all key tables in the Model View.
- **KPI Scorecard:** Built top-level metrics tracking Total Revenue, Active Customers, Average Order Value, and Total Orders.
- **Sales & Operational Visuals:** Developed visuals for monthly sales trends, a geographical map for sales by store, and a staff performance report.
- **Inventory Tracking:** Created a Low Stock Alert dashboard utilizing conditional formatting to highlight items falling below the 10-unit threshold.

- **Customer Insights:** Visualized the Python-generated machine learning segments (e.g., Loyal, At-Risk) using a Donut chart and mapped purchase patterns by geography.
- **Interactivity:** Integrated global slicers allowing end-users to dynamically filter the entire report by Order Status, Product Category, Store, and Customer Segment.

5. Final Deliverables

The project successfully yielded the following assets:

- Fully cleaned, validated Excel datasets.
- A comprehensive SQL relational database with executed query scripts.
- A Python Jupyter Notebook containing documented EDA and RFM segmentation models.
- A highly interactive Power BI dashboard (.pbix file) consolidating ETL, SQL, and ML insights.

6. Conclusion

This retail analytics project successfully bridged the gap between raw transactional data and strategic business intelligence. By executing a seamless ETL pipeline—beginning with robust data standardization in Excel, structured storage in SQL, advanced behavioral modeling in Python, and concluding with a dynamic visualization layer in Power BI—the business is now equipped with a 360-degree view of its operations. Management can now instantly identify top-performing regions, optimize inventory to prevent stockouts, and launch highly targeted marketing campaigns based on data-driven customer segments. The multi-tool approach ensures the solution is scalable, accurate, and directly aligned with the company's growth objectives.